

A whale has finger bones. Why?

Learning intention

Interpret fossil, anatomical and DNA evidence to justify a claim about evolution and state a limitation.

Today: use evidence to write a museum caption.

You may explain by speaking, writing or annotating.

Retrieve the tools

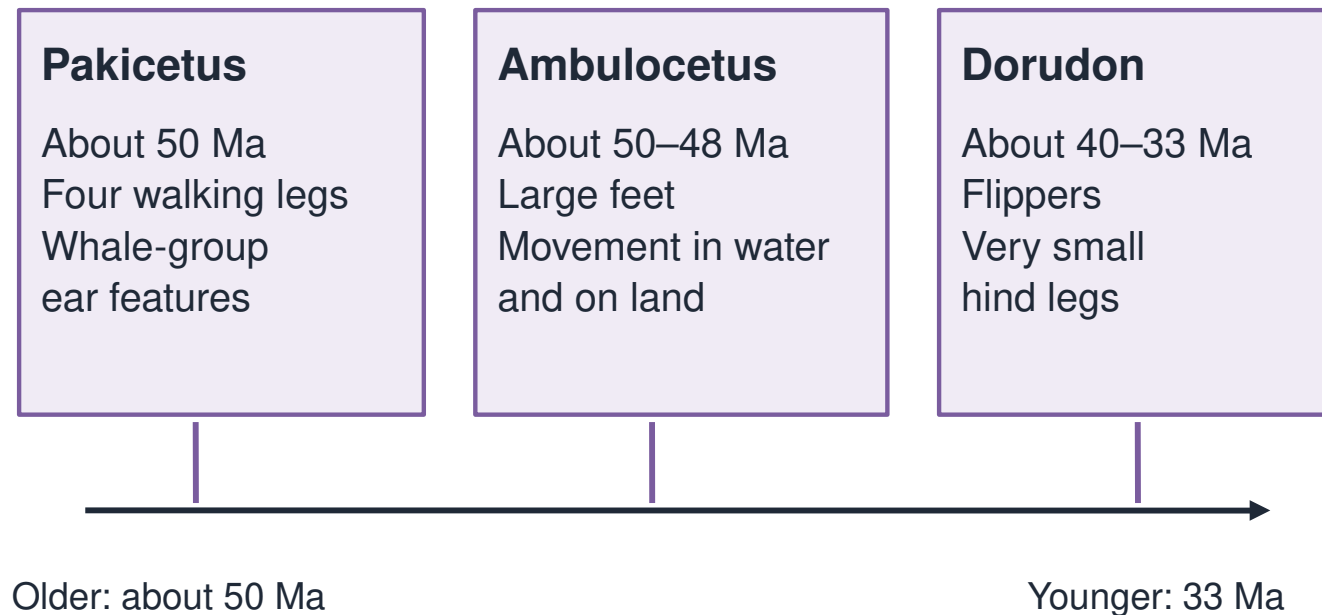
R1. What carries inherited information?

R2. Does evolution describe change in one animal during its life, or populations over generations?

R3. Which is older: 50 Ma or 33 Ma?

Read fossils as evidence

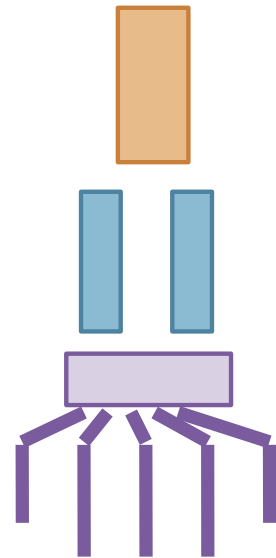
Fossils reveal different combinations of traits



A dated fossil records a past combination of traits. Compare the legs and flippers through time. This timeline does not prove a direct ancestor chain.

The same underlying plan

Human forelimb



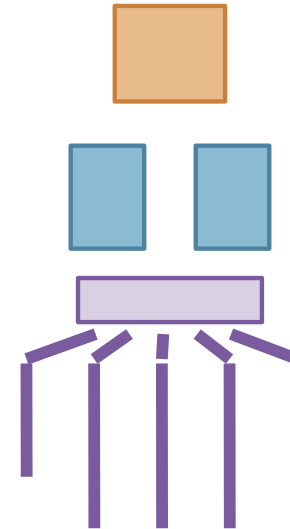
1 upper bone

2 lower bones

Wrist bones

5 digits

Whale flipper



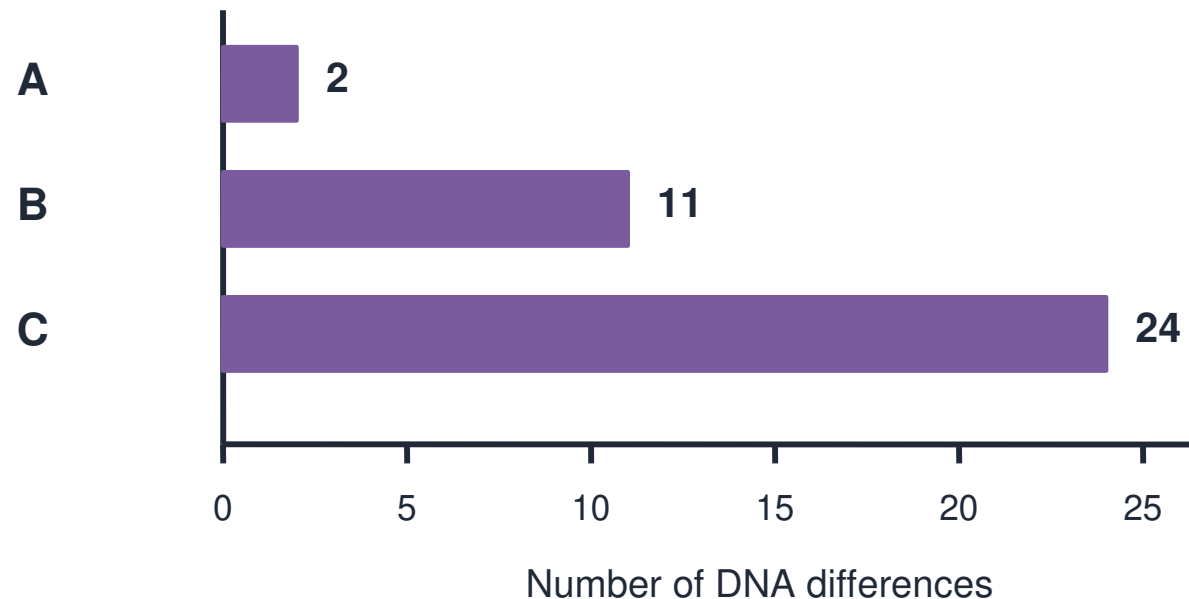
Same bone plan; different shapes. Not to scale.

Compare corresponding bones, not just the outline. The inherited pattern can serve different functions. Shared origin is a stronger claim than “they look alike”.

Read the DNA graph carefully

Constructed sample data: DNA compared with Q

Differences in the same aligned 100 DNA positions

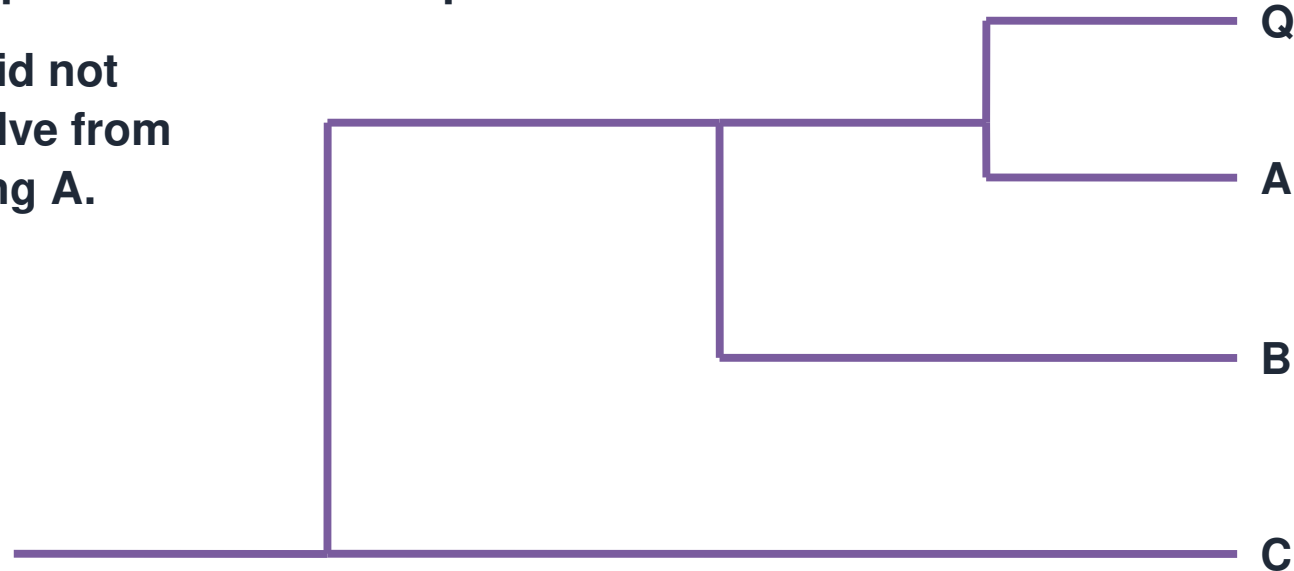


Name the quantity before making a claim. Q and A differ at 2 of the 100 sampled positions. This supports closer relatedness than Q and C.

Evolution branches

One possible relationship model

Q did not
evolve from
living A.



Tips = living species in this model
Branch lengths are not a time scale.

Living species sit at the tips. A shared ancestor sits at a branch point. No living tip is labelled “more evolved”.

Museum label rescue

W1. Read the claim and choose: keep, edit or reject.

“Whales came from living humans because both have similar finger-bone patterns.”

Use Evidence A and F on your card sheet.

Which part is an observation, and which part is an unjustified inference?

A second label arrives

W2. “Q and A are identical. The graph proves A became Q.”

Use the DNA card to repair both errors.

W3. Why use more than one evidence type?

Which claim is best supported?

The sample has 2 Q/A differences and 24 Q/C differences. Choose a claim and be ready to explain one word.

- A** Q evolved directly from living A.
- B** Q and A probably share a more recent ancestor than Q and C.
- C** Q and C share no ancestor.
- D** A and Q have identical genomes.

Create the museum evidence panel

Choose Supported, Standard or Stretch with your teacher.

Explain a claim using two evidence types.

Include one number and one limitation.

A useful explanation structure

The evidence shows ...

This supports ... because ...

It cannot tell us ...

Check your scientific decisions

Have you named an actual feature or number?

Have you explained how it supports the claim?

Have you avoided a claim the evidence cannot test?

Compare, improve, hear the response

Share one evidence sentence in your chosen way.

Hear one helpful question.

Revise one claim that was too strong.

What counts as a strong panel?

Accurate observation plus explained inference.

A number with its sampled quantity.

A specific limit on the conclusion.

One precise final claim

E1. Which species is closest to Q in the sample? Cite a number.

E2. Finish: a shared limb plan supports ...

E3. Name one thing this evidence cannot establish.